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IEGR4320 Data Science in Practice (2025-26)

Assignment 3

Short questions (25%)

Answer all questions.

1. You are developing an AI-driven game agent for Tencent Games' chess platform, utilizing reinforcement learning and decision-making algorithms to enhance strategic gameplay and simulate human-like competitive behavior.

- a. What is reinforcement learning? (2%)

Reinforcement learning(RL) is one of the area of the machine learning, by performing actions in an environment, it can make agent learn to make sequential decisions, to maximize a cumulative reward signal.

- b. Why is reinforcement learning applied for training the game agents? (1%)

As it allows an agent to master complex games, by learning optimal strategies through self play and trial and error, like chess, without the pre-existing data of perfect moves.

- c. How can we categorize reinforcement learning? Explain briefly. (4%)

1. Model-based vs. Model-free:

Model-based agents can learn plan from the model of the environment's dynamics.

Model-free agents learn experience from the policies or value functions.

2. Value-based vs. Policy-based:

Value-based methods derive a policy by learning the value of states

Policy-based methods directly states actions by learning what policy maps.

- d. Suppose you are training a game agent for a chess game, what are the states in the game? Explain your answer with one of the examples below.

(3%)

In this sub-problem, you can decide the chess game (example) as either (i) Chinese chess game, (ii) GO chess game, or (iii) international chess game.

In(iii) international chess game, state is the complete snapshot of any given moment in the game, including every piece's position on the board (what type of piece staying a1-h8), whose turn it is, castling rights, and any possibilities of en passant. This is the board configuration that can represent in number and as input use for a neural network.

e. Based on the same chess game you chose in part (d), what are the actions in the game? Explain your answer with examples. (3%)

Actions in this game are every legal moves make from player in the current state, like from the initial game state, the possible actions from White can include moving pawn (e2 to e4), moving knight (g1 to f3), and about moves from each type of piece(about 18 types) and any position a board (including a1-h8).

- f. Based on the same chess game you chose in part (d), how can you design the value-based reward? Explain your answer with examples. (3%)

Reward is the numerical score assigned after every action.

Simple reward design: terminal reward: +1 when winning, -1 when losing, and 0 for draw, and reward 0 for intermediate moves. There can be a small intermediate rewards, like +0.3 after eat opponent's knight or -0.5 after lose your own bishop.

- g. Based on the same chess game you chose in part (d), how can you design the policy-based reward? Explain your answer with examples. (3%)

The reward function itself is independent to learning method. Policy-based reward signal (like +1 after win) can directly update the agent's policy. Like after win, the probabilities of taking the actions' sequence that lead to victory increase. This policy represents by neural network, which can learn that non-linear functions map states to action probabilities.

- h. When training a game agent as an opponent, will the game agent always learn a better game strategy? Explain briefly. (3%)

No, agent cannot always learn a better strategy, it can get stuck in a sub-optimal policy (a local optimum) and overfit to strategies to training partners, which can make it perform poorer to new strategy.

- i. When training a game agent as an opponent, do we set the goal as training a super skillful and perfect game agent? Explain briefly. (3%)

No, commercial game platform need to create an engaging user experience for player, a super skillful and perfect game agent will make player feel depressed to the game. So, it is better to create adjustable agents with human-like skill levels, that it can provide a more fair and enjoyable challenge to player.